

ORIGINAL ARTICLE

Dual Targeting of Extramedullary Myeloma with Talquetamab and Teclistamab

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ABSTRACT

BACKGROUND

Patients with plasmacytomas that are noncontiguous with bone marrow (true extramedullary myeloma) are at high risk for disease progression or relapse. Phase 1 of the RedirecTT-1 study showed promising efficacy with dual-antigen targeting of myeloma with talquetamab (anti-G protein–coupled receptor family C group 5 member D) plus teclistamab (anti-B-cell maturation antigen) in patients with triple-class–exposed relapsed or refractory multiple myeloma, including those with true extramedullary myeloma.

METHODS

In this phase 2 study, we investigated talquetamab plus teclistamab exclusively in patients with drug-resistant, true extramedullary myeloma. The primary end point was overall response, evaluated with the use of functional imaging. Secondary end points included the duration of response, progression-free survival, overall survival, and safety.

RESULTS

A total of 90 patients were enrolled in the study and received treatment (median follow-up, 12.6 months). A response occurred in 79% of the patients (95% confidence interval [CI], 69 to 87). Among the patients with a response, the percentage with a response duration of at least 12 months was 64% (95% CI, 48 to 76). At 12 months, progression-free survival was 61% (95% CI, 50 to 71), and overall survival was 74% (95% CI, 63 to 83). Common adverse events of any grade included oral symptoms, such as dysgeusia, dry mouth, and dysphagia (in 87% of the patients); cytokine release syndrome (in 78%); and nonrash skin effects (in 69%). Grade 3 or 4 adverse events (most commonly hematologic events) occurred in 76% of the patients; 31% had grade 3 or 4 infection. A nonfatal adverse event led to discontinuation of one or both agents in 6% of the patients. Among 10 deaths that occurred during follow-up, 5 were due to infection and 5 were considered to be related to the study treatment.

CONCLUSIONS

Most patients with drug-resistant, true extramedullary myeloma had a response with talquetamab plus teclistamab. The incidence of adverse events of grade 3 or above was high and was consistent with previous observations for each agent as monotherapy. (Funded by Johnson & Johnson; RedirecTT-1 ClinicalTrials.gov number, NCT04586426.)

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IN MULTIPLE MYELOMA, TRUE EXTRAMEDULLARY disease (i.e., plasmacytomas noncontiguous with bone that do not disrupt the bone cortex) is a high-risk factor for progression, relapse, or both because of its association with therapeutic resistance and evasion of cellular apoptosis.¹⁻³ Approximately 0.5 to 5.2% of patients with myeloma have true extramedullary disease at the time of diagnosis, and this percentage increases to 3.4 to 14.0% in the context of relapsed or refractory disease.^{1,4} Although data are sparse,⁵ outcomes in this patient population are poor as compared with those in patients with paramedullary disease (i.e., plasmacytomas contiguous with bone that extend into soft tissue after cortical disruption).^{2,4,6-9} In a prospective, noninterventional study, median overall survival among patients with extramedullary myeloma treated with standard therapies was 7.2 months and median progression-free survival was 2.6 months.¹⁰ In a meta-analysis, patients with true extramedullary myeloma were 87% less likely to have a response to treatment and approximately twice as likely to have disease progression or die as patients without extramedullary disease⁵; however, few prospective interventional studies have assessed outcomes in large patient populations.

Plasmacytomas exhibit high genomic and micro-environmental complexity, including heterogeneous G protein-coupled receptor family C group 5 member D (GPRC5D) and B-cell maturation antigen (BCMA) expression.^{11,12} Talquetamab (anti-GPRC5D) and teclistamab (anti-BCMA) are first-in-class bispecific antibodies approved as monotherapies for relapsed or refractory myeloma in patients with previous exposure to a proteasome inhibitor, an immunomodulatory drug, and an anti-CD38 monoclonal antibody (triple-class exposure).¹³⁻¹⁷ In the RedirecTT-1 phase 1 study, the recommended phase 2 regimen of talquetamab plus teclistamab elicited a response in 80% of the patients, with a complete response or a stringent complete response in 52%; the probability of a 12-month duration of response was 91%.¹⁸ The observed safety profile was consistent with that of each agent alone in patients with relapsed or refractory myeloma.¹⁸ In a subgroup of patients with true extramedullary myeloma, a response occurred in 61% and the probability of a 12-month duration of response was 82%.¹⁸ These data suggest that dual targeting of GPRC5D and BCMA

with talquetamab and teclistamab may lead to a response in a higher percentage of patients, with greater depth and durability of the response, than targeting either antigen alone. Here, we report the results of the RedirecTT-1 phase 2 study of talquetamab plus teclistamab in patients with true extramedullary myeloma.

METHODS

STUDY DESIGN AND PATIENTS

We conducted a multicenter, nonrandomized, open-label, phase 1b–2 study of talquetamab plus teclistamab in patients with relapsed or refractory multiple myeloma. In the phase 2 cohort, patients had true extramedullary myeloma, defined by at least one nonradiated, bone-independent, soft-tissue plasmacytoma (≥ 2 cm in the greatest dimension) confirmed by central review of whole-body positron-emission tomography and computed tomography (PET-CT) scans; magnetic resonance imaging (MRI) was permitted as an alternative to PET-CT with sponsor approval. Patients with paramedullary plasmacytomas in addition to true extramedullary myeloma were eligible for the study; however, patients with paramedullary plasmacytomas alone, central nervous system involvement, or plasma-cell leukemia were excluded. Patients had confirmed myeloma according to International Myeloma Working Group (IMWG) criteria.¹⁹ Nonsecretory or oligosecretory disease was permitted. Eligible patients had disease that was in relapse or refractory to one or more established therapies, including the most recent line of therapy, or had discontinued such therapies because of unacceptable side effects, and had previous triple-class exposure.

Patients were not permitted to receive radiation therapy beyond the end of cycle 2, and sponsor approval of the use of radiation therapy was required to ensure that response evaluation was not affected. Patients who received radiation therapy to treat a second primary cancer were required to discontinue the study treatment unless they were permitted to continue after consultation with the sponsor and data review. Previous chimeric antigen receptor T-cell (CAR-T) therapy and bispecific antibody therapy were permitted with a therapy-free interval of more than 3 months and more than 21 days, respectively, before beginning study treatment; enrollment of patients

with previous CAR-T therapy was limited to 20% of the study population, and previous bispecific antibodies could not target GPRC5D or BCMA. Previous use of belantamab mafodotin, a BCMA-targeted agent, was permitted. Full eligibility criteria are provided in the protocol, available with the full text of this article at NEJM.org.

STUDY OVERSIGHT

The study was designed by the sponsor, Johnson & Johnson, and was conducted in accordance with the principles of the Declaration of Helsinki and the International Council for Harmonisation guidelines for Good Clinical Practice. The study protocol and amendments were approved by the institutional review board of each site. All the patients provided informed written consent. Data were transcribed by site personnel and were gathered and analyzed by the RedirecTT-1 study group. All the authors vouch for the accuracy and completeness of the data and for the fidelity of the study to the protocol. The RedirecTT-1 study group developed the first draft of the manuscript, with medical writing assistance provided by a professional writer funded by Johnson & Johnson. All the authors reviewed the manuscript, participated in revisions, and agreed to submit the manuscript for publication.

STUDY TREATMENT

Patients received talquetamab subcutaneously at a dose of 0.8 mg per kilogram of body weight plus teclistamab at a dose of 3.0 mg per kilogram every other week in 28-day cycles, after a dose step-up period. Step-up doses were adapted from approved schedules used with each drug as monotherapy. Both bispecific antibodies were administered on the same day, 30 ($\pm$ 10) minutes apart, for all step-up and full treatment doses; additional details are provided in Table S1 in the Supplementary Appendix, available at NEJM.org.

Patients could switch to monthly doses of talquetamab and teclistamab after completing four cycles and having a very good partial response or better or after completing six cycles irrespective of the depth of response, according to investigator discretion. Patients received study treatment until unacceptable side effects developed, consent was withdrawn, disease progression was confirmed, death occurred, or the investigator or sponsor made the decision to discontinue treatment.

END POINTS AND ASSESSMENTS

The phase 2 primary end point was overall response (defined as a partial response or better), assessed by an independent review committee with the use of IMWG 2016 criteria.¹⁹ Assessment included independent central radiology review of whole-body ¹⁸F-fluorodeoxyglucose (FDG) PET-CT scans or, alternatively, diffusion-weighted whole-body MRI results, and findings were interpreted with the use of a functional Deauville score (scores range from 1 to 5, with higher scores indicating greater uptake of FDG at involved sites on PET) and Italian myeloma criteria for PET use (IMPETUS).²⁰⁻²⁴ Because IMWG criteria do not include PET-CT-specific response criteria, adaptations were incorporated to assess extramedullary myeloma response.¹⁹⁻²⁵

A complete response was defined as the disappearance of all plasmacytomas or persistence of fibrotic disease with a Deauville score of 1 to 3, a stringent complete response was defined as a complete response plus a normal free-light-chain ratio and the absence of clonal plasma cells as assessed by immunohistochemistry or flow cytometry, a very good partial response was defined as at least 90% size reduction in all plasmacytomas and reduction in FDG avidity from baseline, and a partial response was defined as at least 50% size reduction in all plasmacytomas and reduction in FDG avidity from baseline.²⁰⁻²⁴ Full response criteria are provided in Table S2. In nonsecretory disease, response was evaluated with the use of functional imaging, and IMWG criteria had to be met for a complete response to be confirmed. Serum and urine samples were obtained every 4 weeks until disease progression. First response was confirmed by PET-CT. Extramedullary myeloma response was assessed at the time of suspected first response and every 12 weeks thereafter until disease progression; imaging frequency could be reduced to every 16 weeks after 1 year.

Secondary end points included time to response, very good partial response or better, complete response or better (i.e., a complete response or a stringent complete response), stringent complete response, the duration of response, progression-free survival, overall survival, and safety. Adverse events, which were graded according to Common Terminology Criteria for Adverse Events, version 5.0, were recorded up to 30 days after the last treatment dose or until the start of subse-

quent antimyeloma therapy, whichever occurred first. Cytokine release syndrome and immune effector cell–associated neurotoxicity syndrome (ICANS) were graded according to the guidelines of the American Society for Transplantation and Cellular Therapy.²⁶

STATISTICAL ANALYSIS

Response, time to response, response duration, progression-free survival, overall survival, and safety were evaluated in all the patients who received at least one treatment dose. Overall response was calculated with corresponding exact 95% confidence intervals. Response duration, progression-free survival, and overall survival were estimated with Kaplan–Meier methods. Across primary and secondary end points, confidence intervals were not adjusted for multiplicity and should not be used to infer treatment effects. Safety was analyzed descriptively with counts and proportions. For overall response, the hypothesis that talquetamab plus teclistamab would have significant

antimyeloma activity in patients with true extramedullary myeloma was tested; the null hypothesis was rejected if the lower bound of the 95% confidence interval exceeded 40%. The sample size was determined on the basis of the primary end point (overall response) and was calculated to provide the trial with 96% power to test the hypothesis; additional details with regard to sample size determination and justification are provided in the protocol. Data were analyzed with the use of SAS software, version 9.4 (SAS Institute).

RESULTS

PATIENTS

Between July 5, 2023, and September 16, 2024, a total of 132 patients underwent screening (Fig. 1). As of March 18, 2025, a total of 90 patients with extramedullary myeloma had received study treatment; the median follow-up was 12.6 months (range, 0.5 to 19.5). Among these 90 patients, 49 (54%) were still receiving treatment at the time of this data-cutoff date for the primary analysis, including 2 patients who were receiving teclistamab only (Fig. 1).

The median age was 64.5 years, and the patients had received a median of four previous lines of treatment over a median duration of 4.7 years since receiving the diagnosis of myeloma (Table 1 and Table S3). The patients had a median of two disease sites (range, one to seven); 54 patients (60%) had at least one soft-tissue site, 35 (39%) had at least one lymph-node site, and 30 (33%) had at least one organ site. The most common disease sites were in the retroperitoneum, abdominal wall, and chest wall (Table S4). In 19 patients (21%), disease at paramedullary sites was present alongside true extramedullary disease. The tumor volume at baseline was less than 25 cm³ in 49 patients (54%), 25 to 50 cm³ in 19 patients (21%), and more than 50 cm³ in 22 patients (24%). Nonsecretory disease was present in 4 patients (4%), and 31 (34%) had oligosecretory disease. Among all the patients, 18 (20%) had previously received anti-BCMA CAR-T therapy and 8 (9%) had previously received bispecific antibodies (all anti-Fc receptor homologue 5). The median time from CAR-T therapy to study treatment was 295 days (range, 98 to 1030). These characteristics were generally representative of the overall population of patients with relapsed or refractory extramedullary myeloma (Table S5).

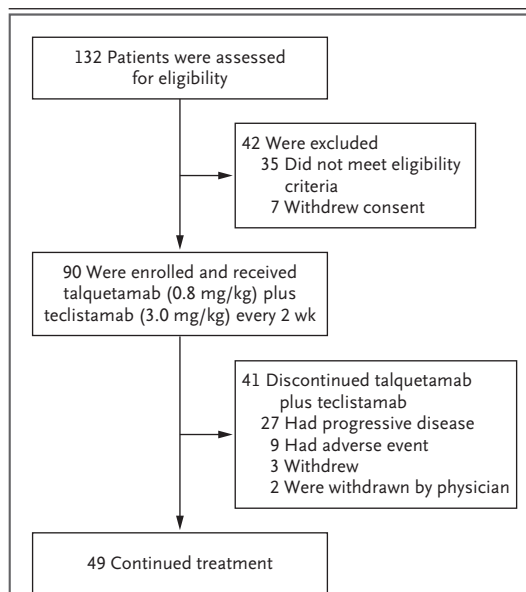


Figure 1. Screening, Enrollment, and Follow-up.

Included among the 49 patients who were receiving ongoing treatment at the time of data cutoff are 2 patients who discontinued talquetamab because of one or more adverse events but continued to receive teclistamab only, as permitted by the study protocol; after beginning treatment on day 1 of the study, 1 patient discontinued talquetamab on study day 10 and 1 patient discontinued talquetamab on study day 79.

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	All Patients (N=90)
Bone marrow plasma cells $\geq 60\%$ — no./total no. (%)†	10/86 (12)
At least 1 true extramedullary plasmacytoma — no. (%)‡§	90 (100)
Extramedullary plasmacytomas‡	
Median no. (range)	2 (1–7)
Distribution — no. (%)	
1	38 (42)
2–3	29 (32)
≥ 4	23 (26)
Extramedullary myeloma tumor volume — no. (%)	
$< 25 \text{ cm}^2$	49 (54)
$25\text{--}50 \text{ cm}^2$	19 (21)
$> 50 \text{ cm}^2$	22 (24)
High cytogenetic risk — no./total no. (%)¶	14/65 (22)
Measurable disease — no. (%)	
Nonsecretory	4 (4)
Oligosecretory	31 (34)
Median time since diagnosis (range) — yr**	4.7 (0.7–21.4)
Previous lines of treatment	
Median no. (range)	4 (1–10)
Distribution — no. (%)	
≤ 3	39 (43)
> 3	51 (57)

* Data are shown for all the patients who received at least one dose of the study treatment. Percentages may not total 100 because of rounding.

† The percentage of plasma cells in bone marrow was assessed with the use of bone marrow biopsy or aspiration in patients with available data.

‡ Patients had at least one nonradiated, bone-independent, soft-tissue plasmacytoma ($\geq 2 \text{ cm}$ in the greatest dimension) confirmed by central review of positron-emission tomography and computed tomography; for 6 patients, the number of extramedullary plasmacytomas was determined by the investigator only.

§ Paramedullary plasmacytomas were also present in 19 patients.

¶ High cytogenetic risk was defined as the presence of a del(17p), t(4;14), or t(14;16) abnormality, as assessed by fluorescence in situ hybridization or karyotype testing.

|| The presence of measurable disease was determined according to International Myeloma Working Group criteria. Nonsecretory disease was defined by the lack of a detectable monoclonal (M)-protein spike in serum or urine, negative results from serum and urine immunofixation electrophoresis, and normal free-light-chain values. Oligosecretory disease was defined by an M-protein spike of less than 1 g per deciliter, with or without positive results from serum immunofixation electrophoresis; a urine M-protein level of less than 200 mg per 24 hours as determined by electrophoresis, with or without positive results from urine immunofixation electrophoresis; and an involved serum free-light-chain level of less than 10 mg per deciliter.

** The time since diagnosis is reported for 89 patients with available data.

EFFICACY

A response occurred in 71 of 90 patients (79%; 95% confidence interval [CI], 69 to 87); this result was substantially greater than the hypothesized 40%. A very good partial response or better occurred in 63 patients (70%; 95% CI, 59 to 79), and a complete response or better in 49 (54%;

95% CI, 44 to 65) (Fig. 2A). Among 18 patients who had previously received anti-BCMA CAR-T therapy, 15 (83%; 95% CI, 59 to 96) had a very good partial response or better and 14 (78%; 95% CI, 52 to 94) had a complete response or better. Among 8 patients who had previously received bispecific antibody therapy, 6 (75%; 95% CI, 35 to

97) had a response; 5 patients (62%; 95% CI, 24 to 91) had a very good partial response or better, and 3 (38%; 95% CI, 8 to 76) had a complete response or better (Fig. 2A).

The median time to first response was 2.6 months (range, 1.0 to 5.8), and the median time to best response was 4.7 months (range, 1.0 to 11.9). Responses were similar across clinically relevant subgroups, including patients with more than three previous lines of therapy, International Staging System stage III myeloma, four or more extramedullary sites at baseline, or high-risk cytogenetics (Fig. S1). Among patients with organ disease sites, 83% (95% CI, 65 to 94) had a response, as did 77% of the patients (95% CI, 64 to 87) with nonorgan disease sites. The overall response was 82% (95% CI, 68 to 91) among patients with a tumor volume of less than 25 cm² at baseline, 74% (95% CI, 49 to 91) among those with a tumor volume of 25 to 50 cm², and 77% (95% CI, 55 to 92) among those with a tumor volume greater than 50 cm².

The median response duration was 13.8 months (95% CI, 11.5 to not estimable); among the patients with a response, the percentage with a response duration of at least 12 months was 64% (95% CI, 48 to 76) (Fig. 2B). In patients with previous CAR-T therapy, the median response duration was 13.8 months (95% CI, not estimable) and 72% (95% CI, 41 to 89) had a response duration of at least 12 months. At the time of data cutoff, 66% of the patients with a response were continuing to receive any study treatment (median follow-up, 13.4 months; range, 1.2 to 19.5). Among 71 patients with a response, 55 (77%) switched to monthly dosing (Fig. S2). Most responses (51 of 55 [93%]) deepened or were maintained in the first 6 months after the switch to monthly dosing.

Median progression-free survival was 15.4 months (95% CI, 10.8 to not estimable); 12-month progression-free survival was 61% (95% CI, 50 to 71) (Fig. 2C). In patients with previous anti-BCMA CAR-T therapy, median progression-free survival was 15.4 months (95% CI, 4.9 to not estimable) and 12-month progression-free survival was 61% (95% CI, 35 to 79). Median overall survival was not yet determined; 12-month overall survival was 74% (95% CI, 63 to 83) (Fig. S3). After treatment with talquetamab plus teclistamab, 22 patients (24%) received at least one subsequent antineoplastic therapy.

SAFETY

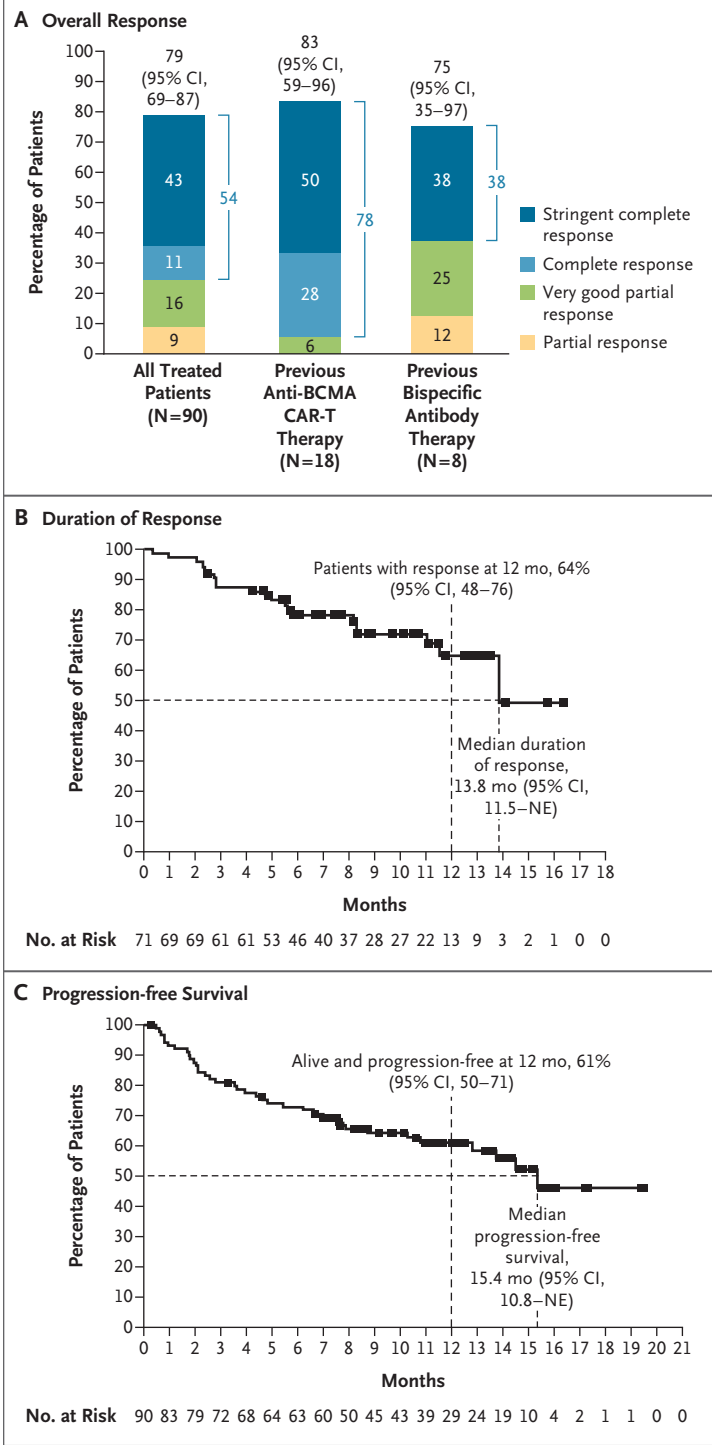
The most common adverse events of any grade were oral symptoms, cytokine release syndrome, and nonrash skin events (Table 2). The most common adverse events of grade 3 or 4 were cytopenias; these events were generally transient and resolved in most patients (87%). Treatment modification due to adverse events occurred in 62 patients (69%); in 27 patients (30%), treatment was modified owing to infection. Treatment cycles were delayed because of adverse events in 55 patients (61%). Nonfatal events led to discontinuation of one or both agents in 5 patients (6%): both talquetamab and teclistamab were discontinued by 1 patient with grade 4 ICANS, 1 patient with dysphagia and dry mouth, and 1 patient with pseudomonal pneumonia and pseudomonal sepsis, and talquetamab only was discontinued by 1 patient with dysgeusia and dysphagia and 1 patient with hypohidrosis. Of these patients, 3 (3%) had events that were deemed by the investigator to be related to the study treatment (Table S6). No patients discontinued teclistamab only.

Adverse events resulted in death in 10 patients (11%), 7 of whom had stable disease or disease progression at the time of death. Five deaths were deemed by the investigator to be treatment related (one death due to coronavirus disease 2019 [Covid-19]-related pneumonia, one due to klebsiella pneumonia, one due to pneumonia caused by unspecified infection, one due to pseudomonal sepsis, and one due to aspiration), and five were not considered to be treatment related (one death due to klebsiella sepsis, one due to cerebral hemorrhage, one due to euthanasia, one due to general deterioration of physical health, and one due to respiratory failure) (Table S7).

Cytokine release syndrome occurred in 70 patients (78%). All the cases were grade 1 or 2, most were confined to the step-up period and the first cycle of doses, and almost all (121 of 122 [99%]) were resolved at the time of data cutoff. Of these 70 patients, 68 (97%) received supportive measures for cytokine release syndrome (Table S8). The median time from the most recent treatment dose to the onset of cytokine release syndrome and the median duration of the syndrome were 2 days each. ICANS occurred in 11 patients (12%). Most cases were grade 1 or 2, with one grade 3 and one grade 4 event reported; the grade 4 event had resolved by the data-cutoff date. Of these 11 patients with ICANS, 10 (91%)

Figure 2. Response and Kaplan–Meier Analysis of Progression-free Survival with Talquetamab plus Teclistamab in Patients with True Extramedullary Myeloma.

Panel A shows percentages of patients with a response confirmed by an independent review committee according to International Myeloma Working Group 2016 criteria, with adaptations for extramedullary myeloma response as assessed by functional imaging, among all the patients with relapsed or refractory multiple myeloma with extramedullary disease who received study treatment (talquetamab at a dose of 0.8 mg per kilogram of body weight plus teclistamab at a dose of 3.0 mg per kilogram every other week). Percentages within each group may not add up to the total for the group because of rounding. Overall response was assessed as a stringent complete response, a complete response, a very good partial response, or a partial response. The brackets indicate the percentage of patients with a complete response or a stringent complete response. A stringent complete response was defined as a complete response plus a normal free-light-chain ratio and the absence of clonal plasma cells as assessed by immunohistochemistry or flow cytometry. Panel B shows the duration of response in all the patients who received treatment and had a confirmed partial response or better. The duration of response was calculated as the number of months from the first documented response to the earliest date of progression or death from any cause; data for patients who were living and free of progression were censored at the date of the most recent disease assessment. Boxes indicate censored data; data for 50 patients (70%) were censored at the data-cutoff date. The dashed vertical lines indicate the percentage of patients with a response at 12 months and the median duration of response. The dashed horizontal line indicates 50% of the patients. Panel C shows progression-free survival from the beginning of treatment among all the patients who received treatment. Events were assessed according to the earliest date of progression or death from any cause; data for patients who were living and free of progression were censored at the date of the most recent disease assessment. Data for patients without disease assessments were censored with 1 day of follow-up. Boxes indicate censored data; data for 53 patients (59%) were censored at the data-cutoff date. The dashed vertical lines indicate progression-free survival at 12 months and median progression-free survival. The dashed horizontal line indicates 50% of the patients. Confidence intervals were not adjusted for multiplicity and should not be used to infer treatment effects. BCMA denotes B-cell maturation antigen, CAR-T chimeric antigen receptor T-cell, and NE not estimable.



received supportive measures (9 received glucocorticoids) (Table S9). All the ICANS events occurred during the step-up period or the first cycle of doses, and the median duration was 2 days.

Oral adverse events occurred in 78 patients

(87%) and were generally low grade, with four grade 3 or 4 events (Table S10). Dysgeusia occurred in 63 patients (70%), dry mouth in 40 (44%), and dysphagia in 26 (29%). All these events were low grade except for grade 3 dysphagia events

Table 2. Hematologic and Nonhematologic Adverse Events, Excluding Infections.*

Event	Any Grade	Grade 3 or 4
	<i>no. of patients (%)</i>	
Any adverse event	90 (100)	68 (76)
Hematologic events		
Neutropenia	65 (72)	56 (62)
Anemia	46 (51)	28 (31)
Thrombocytopenia	34 (38)	23 (26)
Nonhematologic events		
Oral adverse event†	78 (87)	4 (4)
Cytokine release syndrome	70 (78)	0
Nonrash skin-related adverse event‡	62 (69)	0
Nail-related adverse event§	50 (56)	0
Weight decrease	48 (53)	10 (11)
Dry mouth	40 (44)	0
Cough	33 (37)	0
Diarrhea	30 (33)	3 (3)
Pyrexia¶	28 (31)	1 (1)
Hypokalemia	27 (30)	7 (8)
Fatigue	27 (30)	3 (3)
Nausea¶	27 (30)	0

* Adverse events listed are those that occurred in at least 30% of the total study population. Adverse events that emerged during treatment up to 30 days after the last treatment dose were recorded. Patients could have multiple adverse events. Adverse events were graded according to Common Terminology Criteria for Adverse Events, version 5.0. None of the most common adverse events reported were grade 5 (events that resulted in death).

† Oral adverse events included ageusia, cheilitis, dry mouth, dysgeusia (maximum severity is grade 2 according to Common Terminology Criteria for Adverse Events), dysphagia, glossitis, glossodynia, hypogeusia, mouth ulceration, oral discomfort, oral mucosal erythema, oral pain, stomatitis, swollen tongue, taste disorder, tongue discomfort, tongue edema, tongue erythema, and tongue ulceration.

‡ Nonrash skin-related adverse events included skin exfoliation, dry skin, pruritus, and palmar-plantar erythrodysesthesia syndrome.

§ Nail-related adverse events included nail discoloration, nail disorder, nail dystrophy, nail ridging, nail toxicity, onychoclasia, onycholysis, and onychomadesis.

¶ Symptoms of cytokine release syndrome or immune effector cell–associated neurotoxicity syndrome were excluded from the events reported here.

in 2 patients (2%); one resolved after 11 days and one was unresolved at the data-cutoff date. Nonrash skin events occurred in 62 patients (69%), and nail-related events in 50 patients (56%); all these events were low grade. Rash-related events occurred in 26 patients (29%), and almost all were low grade.

Infections occurred in 71 patients (79%), and grade 3 or 4 infections in 28 patients (31%) (Ta-

ble 3). The onset of grade 3 or 4 infection was most common in the first 6 months of treatment, and the incidence of infection then declined (Table S11). Of the 5 patients who died from infections, 3 did not receive intravenous immune globulin and 3 had IgG levels of less than 400 mg per deciliter at the time of death. Antiherpes prophylaxis was given to 87 patients (97%), and 5 (6%) received antiviral hepatitis B prophylaxis. A Covid-19 vaccine was administered to 74 patients (82%). Opportunistic infections occurred in 6 patients (7%) (Table 3) and were generally confined to the first 6 months of treatment; none were fatal.

A total of 78 patients (87%) received any intravenous immune globulin during the study. At baseline, 20 patients (22%) had IgG levels of less than 400 mg per deciliter; post-treatment, 63 patients (70%) had an investigator-assessed adverse event of hypogammaglobulinemia or IgG levels of less than 400 mg per deciliter. Among these 63 patients, 54 (86%) received at least one dose of intravenous immune globulin, with a median time from the first dose of study treatment to the first dose of intravenous immune globulin of 50 days.

DISCUSSION

This phase 2 study prospectively assessed response by central radiology review, with FDG PET-CT imaging graded according to the Deauville scale and IMPETUS criteria.^{20-22,24} Talquetamab plus teclistamab led to an overall response of 79% (54% of the patients had a complete response or better), and 64% of the patients with a response had a response duration of at least 12 months. The study population included patients with nonsecretory or oligosecretory disease; such patients have historically been excluded from most clinical studies because of restrictive eligibility criteria.²⁷ Our findings offer valuable data for treatment in a population that is both clinically challenging and underrepresented in existing clinical studies. The definition of true extramedullary myeloma in our study is aligned with the emerging consensus that the diagnosis should include only soft-tissue plasmacytomas that are noncontiguous with bone.^{1,4,9}

At 12 months, progression-free survival was 61% and overall survival was 74%, results that compare favorably with durability and survival

outcomes observed with other approved T-cell–redirecting therapies.^{14,28,29} Approximately three quarters of the patients had grade 3 or 4 adverse events, a finding that is consistent with the known safety profile of each agent alone, yet only 6% of the patients discontinued treatment with one or both agents because of a nonfatal adverse event. Adverse events resulted in death in 10 patients, 5 of whom died from infections. Among patients with a response, the response was maintained after a switch to a monthly dosing schedule.³⁰

Cross-study comparisons should be interpreted with caution, particularly because definitions of extramedullary myeloma vary between studies and patient numbers within studies are generally small. The overall response with talquetamab plus teclistamab in this study (79%; 95% CI, 69 to 87) was higher than the overall response observed with standard treatments in previous studies involving patients with triple-class–exposed relapsed or refractory myeloma with a combination of true extramedullary and paramedullary disease (approximately 24%; 95% CI, 10 to 44).¹⁰ In previous studies involving such patients who received T-cell–redirecting therapies, the overall response was 44% (95% CI, 34 to 53) with talquetamab monotherapy and 43% (95% CI, 30 to 58) with teclistamab monotherapy (Johnson & Johnson, unpublished data) and 39% with elranatamab, a BCMA- and CD3-bispecific antibody.³¹ With ciltacabtagene autoleucel, the overall response was 57% (95% CI, 34 to 78) among patients with true extramedullary myeloma, although not all patients had triple-class exposure (Johnson & Johnson, unpublished data), and the overall response with idecabtagene vicleucel was 56% among patients with a combination of true extramedullary and paramedullary myeloma.²⁹

In the current study, responses with talquetamab plus teclistamab were deep (with a complete response or better in 54% of patients). Among patients with extramedullary myeloma, a complete response or better has been observed in 15% of patients with talquetamab monotherapy and in 19% with teclistamab monotherapy (Johnson & Johnson, unpublished data). With standard therapies, 3% of patients with extramedullary myeloma had a complete response or better.¹⁰ Responses to talquetamab plus teclistamab in this study were also similar across clinically relevant subgroups, including patients who had received previous T-cell–redirecting therapies: among those who

Table 3. Most Common Infections.*

Infection	Any Grade	Grade 3 or 4	Grade 5†
	number of patients (percent)		
Any infection	71 (79)	28 (31)	0
Upper respiratory tract infection	22 (24)	3 (3)	0
Covid-19	20 (22)	5 (6)	1 (1)‡
Pneumonia	16 (18)	4 (4)	1 (1)
Urinary tract infection	12 (13)	3 (3)	0
Viral upper respiratory tract infection	9 (10)	2 (2)	0
Oral candidiasis	6 (7)	0	0
Sinusitis	5 (6)	2 (2)	0

* Infections that emerged during treatment up to 30 days after the last treatment dose and occurred in at least 5% of the total study population are reported. Patients could have multiple infections. Among 242 infections, the median time to onset from the last treatment dose was 12 days (range, 1 to 87), and 209 infections (86%) had resolved or the patient had recovered by the data-cutoff date. Among 218 infections with both start and end dates available, the median duration was 13 days (range, 2 to 140). A total of 6 patients (7%) had opportunistic infections, including reactivation of cytomegalovirus infection (in 4 patients), cytomegalovirus infection (in 2 patients), cytomegalovirus esophagitis (in 1 patient), esophageal candidiasis (in 1 patient), and polyomavirus viremia (in 1 patient). Covid-19 denotes coronavirus disease 2019.

† A total of 5 patients died from infections (1 from pneumonia due to Covid-19, 1 from pneumonia due to klebsiella infection, 1 from sepsis due to klebsiella infection, 1 from pneumonia due to unspecified infection, and 1 from pseudomonas sepsis).

‡ The patient died from Covid-19–related pneumonia.

had received anti-BCMA CAR-T therapy, 83% had a response, and among those who had received previous bispecific antibodies (all anti-Fc receptor homologue 5), 75% had a response.

Median progression-free survival was 15.4 months with talquetamab plus teclistamab among patients in this study; in previous studies, median progression-free survival was less than 3 months with approved standard therapies¹⁰ and less than 6 months with approved bispecific antibody monotherapies.⁴ The longer progression-free survival with talquetamab plus teclistamab suggests that this new, off-the-shelf, bispecific antibody combination may be a promising therapeutic option for patients with relapsed or refractory extramedullary myeloma.

The adverse events associated with talquetamab plus teclistamab, including cytokine release syndrome and ICANS, were generally consistent with the known profiles of each agent as monotherapy^{13-15,32} and were not exacerbated with combination therapy. GPRC5D-associated adverse events were common, which is consistent with

the known profile of GPRC5D-targeting therapies, including talquetamab monotherapy.^{13,14,33-36} Events were managed according to protocol recommendations, which included glucocorticoid rinses and nutrition consultation for oral adverse events.^{14,37} Consistent with the results of the phase 1 study, infections of grade 3 or 4 in this phase 2 study occurred mostly in early treatment cycles, when the majority of the patients were receiving treatment every other week; however, the overall incidence in phase 2 was lower than that in phase 1 (31% vs. 64%), and a higher percentage of patients in this study than in the phase 1 study received at least one dose of intravenous immune globulin (87% vs. 57%),¹⁸ findings that highlight the importance of vigilant management of infection.^{38,39}

Limitations of the study include the short duration of follow-up and the single-group, open-label nature of the study. In addition, certain subgroups of patients, such as those with central nervous system involvement or plasmacell leukemia, were excluded. MonumenTAL-6 (ClinicalTrials.gov number, NCT06208150) is an ongoing phase 3 trial assessing talquetamab plus teclistamab in patients with relapsed or refractory myeloma with or without extramedullary disease.⁴⁰

In this study, treatment with talquetamab plus teclistamab in patients with extramedullary myeloma had clinically meaningful efficacy. The treatment was associated with grade 3 or 4 adverse events in 76% of the patients, and some patients died from adverse events related to the study treatment. No new adverse events were noted, and just 6% of the patients discontinued treatment with one or both agents because of nonfatal adverse events. These results show clinical benefit from this off-the-shelf, bispecific antibody combination and validate dual targeting of GPRC5D and BCMA with talquetamab plus teclistamab as a potential therapeutic approach in myeloma.

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